AI 321: Machine Learning for Sequential Data

Introduction to Time-Series Analysis

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Overview

1. Time Series Analysis

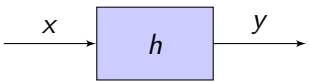
2. Mathematical Notations of Linear Systems

3. Review of Linear Systems
   - Review of Linear Systems
   - Block Diagrams as System Representations
   - Response of LTI systems
   - State-Space Representations

What is time series analysis? [1]

- time series is sequence of observations in time
- sequential data can be found everywhere
- interest in the dependence of observations
- What are the techniques for the analysis of the dependence
- requires stochastic and dynamic models of time series data (modeling)
- Analysis and modeling of the system that generated the outputs (system identification)

Systems Modeling



- h: linear or nonlinear
- h: univariate or multivariate
- stochastic h: stationary or non-stationary
- h: time-invariant or dynamic systems

- Focus: linear, univariate and multivariate, stationary (weak sense), time-invariant and some dynamic systems.

Application areas

- *Forecasting* of future values
- Determination of the *transfer function* of a system subject to inertia
- Assess the effects of unusual *intervention* on the behavior of a time series
- Interrelationship of time-series variables in a *multivariate* system
- Design of simple *control schemes*, to steer the output to the desired target

Forecasting

- Observations at time t to predict future observations at time $t + 1$
- Associate a measure of accuracy to the forecast, probability limits
- Stochastic modeling of the system
  - model identification or specification
  - estimation of model parameters
  - model checking and diagnostics

Estimation of Transfer function

- To better understand the operations of the system
- Determine a hypothetical *impulse response* or transfer function
- Modeling based on deterministic response maybe insufficient in the presence of noise
- Can also be used in forecasting

Analysis of Effects of Unusual Interventions

- Unusual intervention events: new gov't regulations, economic policy changes, strikes, special promotion campaigns
- The effect can be 0 or 1 at the input
- Does the intervention have an impact on the time-series?
- To ensure that forecast and system modeling are not significantly affected by the unusual intervention events.

Analysis of Multivariate Time-Series

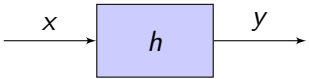
- Many time-series data are available in several interrelated variables
- Analysis is more effective by considering the time-series jointly
- Develop statistical models to understand the dynamic relationship among several time-series

Discrete control schemes

- Process control
- Feedback and feed forward control
- Compensate on the effects of drift and disturbance due to noise
- outside the scope of this course.

Familiar notations for signals and systems

Signal and systems as functions



- $x(t),\ t \in \mathbb{R}$, continuous-time i.e $\sin(2\pi t)$
- $x(n), x(k),\ n, k \in \mathbb{Z}$, discrete-time i.e $x(n) = 2n = \{0, 2, 4, 6, \ldots\}$
- $h(t)$, impulse response
- For linear time-invariant systems (LTI), $h(t)$ and $h(n)$ completely describes the system

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

Signals as Functions [2]

a signal is a function of an independent variable, the independent variable is usually time (t), or $[n]$

- $f(t)$ is a force on some mass
- $v_{out}(t)$ is the output voltage of a circuit
- $p(t)$ is the acoustic pressure on some point
- $x[n]$ is the amount of income at month n

Function Domain and Range

A function is a **mapping** or a **relationship** between two sets of quantities

$$f : X \to Y$$

The **domain** is the set of all allowable values of its input argument. The **range** is the set of all possible output values of the function.
Some common domains:

- all values of t: $t \in \mathbb{R}$
- non-negative values of t: $t \geq 0$
- t in some interval: $a \leq t \leq b$
- t at uniformly sampled points: $t = t_0 + \Delta tk, k = 0, \pm 1, \pm 2, ...$
- *discrete-time signals* are defined for integer t: $t = 0, \pm 1, \pm 2, ...$

Systems as Functions

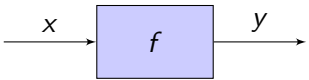
- A system transforms *input signals* into *output* signals
- A system is a *function* mapping input signals into output signals (which are themselves functions)
- we focus on systems with one-input and one-output (SISO)

Notation:

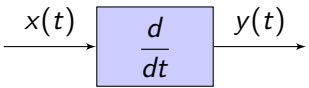
- $y = Hx$ or $y = H(x)$ or $y(t) = H[x(t)]$ means that the system H acts on input signal x to produce output signal y.
- In general, $y = Hx$, does **not** mean multiplication!

Block Diagram representation of Systems

A block diagram represents a **system**, or a function in signal space. Here x and y are variables representing signals in the function space, and f is the system defined as: $y = f(x)$.



Example: Signal derivatives, $y(t) = \dfrac{dx(t)}{dt}$



Note: Algebraic equations are *read* from right to left, block diagrams are from left to right.

Other Mathematical Operators (Box & Jenkins)[1]

- Backward shift operator, B

$$Bz_t = z_{t-1}$$

$$B^m z_t = z_{t-m}$$

- Forward shift operator, $F = B^{-1}$

$$Fz_t = z_{t+1}$$

$$F^m z_t = z_{t+m}$$

- Backward difference operator, ∇

$$\nabla z_t = z_t - z_{t-1} = (1 - B)z_t$$

Examples

- Suppose μ is a parameter (scalar) that determines the "level" or average value of a time-series
- We define an operator based B on a sequence of weights: $\varphi_1, \varphi_2, \ldots$

$$\varphi(B) = 1 + \varphi_1 B + \varphi_2 B^2 + \cdots$$

- Random sequence drawn from a distribution (IID)

$$a_t, a_{t-1}, a_{t-2}, \ldots$$

- We can define a time-series, z_t, as

$$\begin{aligned} z_t &= \mu + a_t + \varphi_1 a_{t-1} + \varphi_2 a_{t-2} + \cdots \\ &= \mu + \varphi(B) a_t \end{aligned}$$

Linear Systems (Univariate)

Definition (Linear systems)

A system **F** is **linear** if the following two properties hold:

1. **homogeneity**: if x is a signal and a is scalar, then

$$F(ax) = aF(x)$$

2. **superposition**: if x_1 and x_2 are any two signals, then

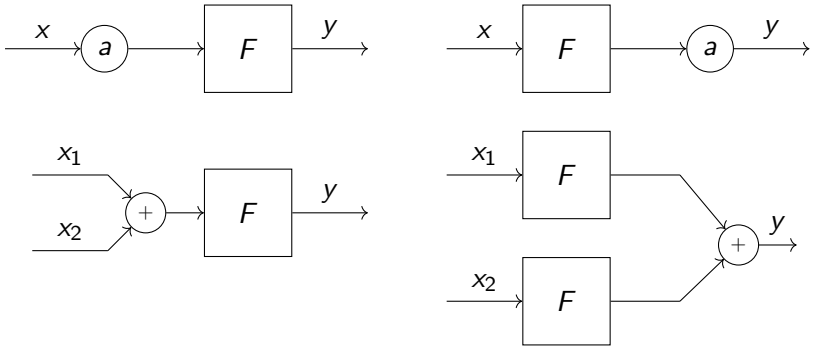
$$F(x_1 + x_2) = F(x_1) + F(x_2)$$

Linearity means:

- scaling before or after the system is the same
- summing before or after the system is the same

Linear property

Linearity means the following pairs of block diagrams are equivalent, they have the same output for any input.



Linear Systems

- **Examples of linear systems:** scaling systems, differentiators, integrators, running average, time shift, convolution, summing, difference systems.
- **Examples of non-linear systems:** sign detector, multiplier, comparator, modulation.

Time-Invariant Systems

Definition (Time-Invariant)

A system is **time-invariant** if delaying or advancing an input will correspondingly delay or advance the output of the system.

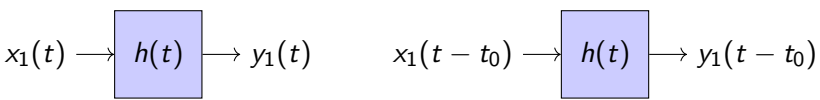
If $\mathbf{H}[\cdot]$ is a time-invariant system, described by $h(t)$, and

$$y_1(t) = H[x_1(t)]$$

then

$$y_1(t - t_0) = H[x_1(t - t_0)]$$

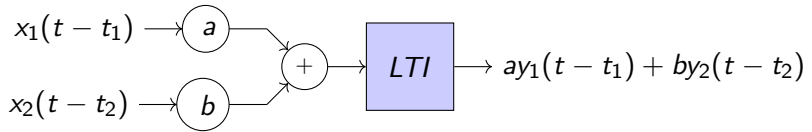
Block diagram: Time-invariant system

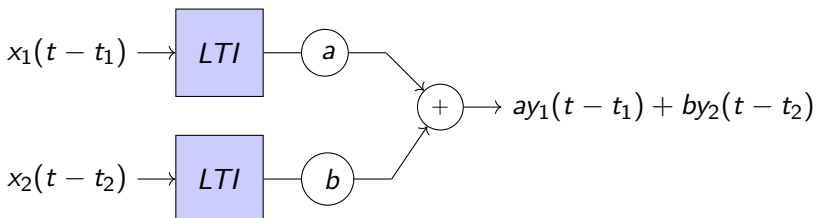


Linear Time-Invariant Systems (LTI)

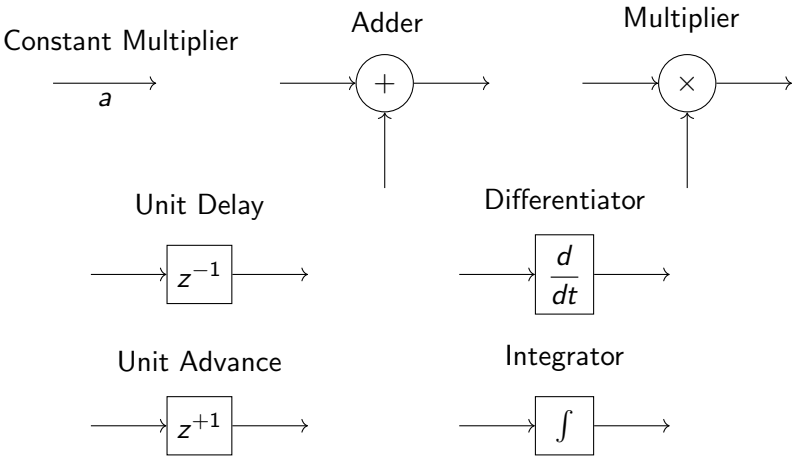
Definition (Linear Time-Invariant)

Linear time-invariant (LTI) systems are both **linear** and **time-invariant**.



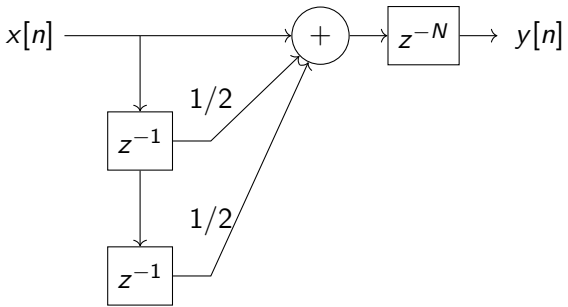


Block diagram elements



Example of Block diagrams of systems

A simple block diagram of a synthesize plucked string guitar sound (Karplus-Strung Algorithm).



Systems described by differential equations (CTS)

Systems described by $y(t) = \mathbf{H}[x(t)]$ as a *linear constant coefficient differential equation* (LCCDE).

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \ldots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \frac{d^m x(t)}{dt^m} + b_{m-1} \frac{d^{m-1} x(t)}{dt^{m-1}} + \ldots + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

Systems described by differential equations (CTS)

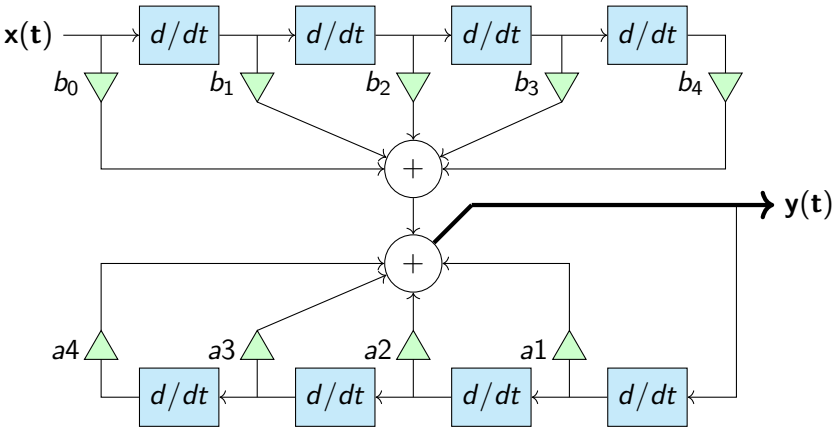
Expressed in a more compact form:

$$\sum_{k=0}^{n} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{m} \frac{d^k x(t)}{dt^k}$$

with given initial conditions $y^{(n-1)}(0), y^{(n-2)}(0), ..., y''(0), y'(0), y(0)$.
where

- n is the order of the system
- $b_m, ..., b_0$ and $a_m, ..., a_0$ are the *coefficients* of the system
- when the initial conditions are zero, the LCCDE becomes **linear**.

System Representation of CTS



Systems described by difference equations (DTS)

Systems described by $y(n) = \mathbf{H}[x(n)]$ are discrete time systems, described by a *difference equation*.

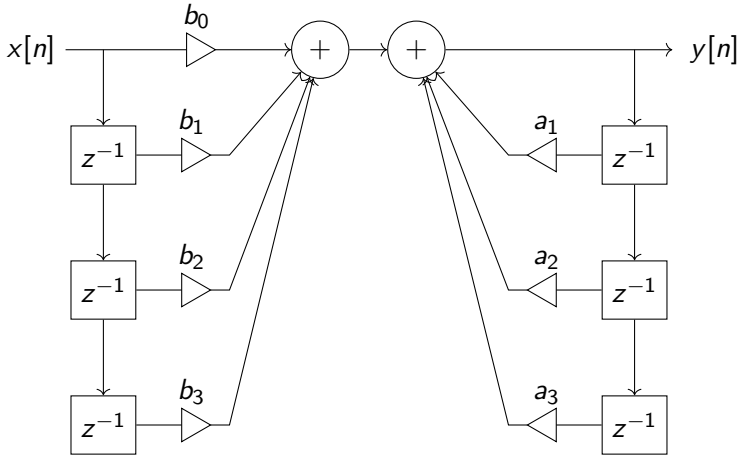
$$a_N y(n) + a_{N-1} y(n-1) + ... + a_1 y(n-1) + a_0 y(n) = \\ b_M x(n-M) + b_{M-1} x(n-M-1) + ... + b_0 x(n)$$

with initial conditions

$$y(k) = y_k, ..., y(-2) = y_2, y(-1) = y_1.$$

System representation of DTS

Direct From I. Direct implementation of the difference equation (AR, MA and ARMA systems)

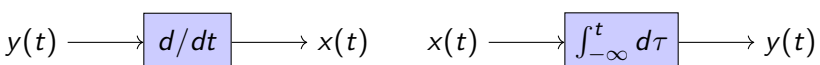


Differential Equations and Difference Equations

1st order $x(t) = \left.\frac{dy(t)}{dt}\right|_{t=nT}$

where n is some integer and T is a time interval called period.

Can be implemented using *integrators*.

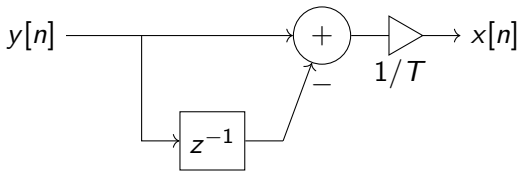


Differential Equations and Difference Equations

1st order

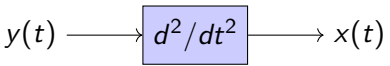
$$x(t) = \left.\frac{dy(t)}{dt}\right|_{t=nT} = \frac{y(nT) - y(nT - T)}{T} = \frac{y(n) - y(n-1)}{T} = x(n)$$

Block diagram



Differential Equations and Difference Equation

2nd order $x(t) = \dfrac{dy^2(t)}{dt^2}$



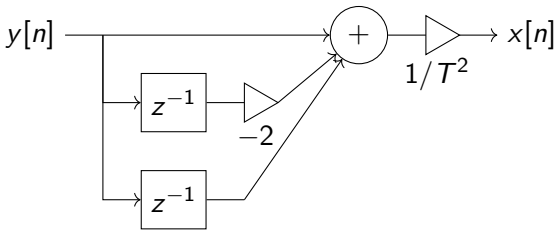
Differential Equations and Difference Equations

2nd order: $x(n) = \left.\frac{dy^2(t)}{dt^2}\right|_{t=nT} = \left.\frac{d}{dt}\left(\frac{dy(t)}{dt}\right)\right|_{t=nT}$

$$x(n) = \frac{\frac{y(nT)-y(nT-T)}{T} - \frac{y(nT)-y(nT-2T)}{T}}{T}$$

$$x(n) = \frac{y(nT) - 2y(nT-T) + y(nT-2T)}{T^2}$$

Block diagram

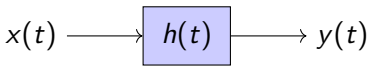


Convolution in CT and DT Systems

Convolution

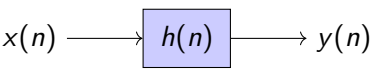
CTS: For system **F** described by $h(t)$, the output $y(t)$ is computed from the convolution integral: $y(t) = x(t) * h(t)$

$$y(t) = \int_{-\infty}^{\infty} h(t-\tau)x(\tau)d\tau$$



DTS: For a system **F** described by $h(n)$, the output $y(n)$ is computed by the convolutional sum: $y(n) = x(n) * h(n)$

$$y(n) = \sum_{k=-\infty}^{\infty} h(n-k)x(k)$$

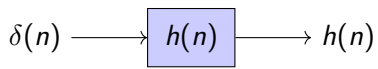


Impulse response of a system

Definition (Impulse response)

The **impulse response** of a system is the output of the system when the input is an **impulse** $\delta(n)$.

The **impulse** is modeled as the *Dirac* delta function $\delta(t)$ in CTS, and modeled as the *Kronecker* delta $\delta(n)$ in DTS.
A linear time-invariant system (LTI) is completely described by its impulse response.



For any LTI system, the output $y(n)$ is calculated from its input $x(n)$ and impulse response $h(n)$ via **convolution** operation.

Linear Systems (Multivariate) [3]

- a continuous-time linear dynamical system has the form

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t),$$

$$\mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)\mathbf{u}(t)$$

  where
  - $\mathbf{x}(t) \in \mathbb{R}^n$: (n x 1) state vectors
  - $\mathbf{u}(t) \in \mathbb{R}^m$: (m x 1) vector of input functions
  - $\mathbf{y}(t) \in \mathbb{R}^p$: (p x 1) vector of output
  - $\mathbf{A}(t) \in \mathbb{R}^{nxn}$: (n x n) dynamics or state matrix
  - $\mathbf{B}(t) \in \mathbb{R}^{nxm}$: (n x m) input matrix
  - $\mathbf{C}(t) \in \mathbb{R}^{pxn}$: (p x n) output or sensor matrix
  - $\mathbf{D}(t) \in \mathbb{R}^{pxm}$: (p x m) feed through matrix
- the state-space representation of a system
- A continuous-time system is first order vector *differential equation*.

State Space Representation of Linear Systems

- For conciseness the equations are often written as:

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

- Most linear systems are time-invariant. A, B, C and D are constants, not dependent on t
- When there is no input u (hence, no B or D), the system is called autonomous
  very often there is no feed through, i.e $D = 0$
- sometimes called "m-input, n-state, p-output" system.

Discrete-time Linear System

- A discrete-time linear dynamical system has the form:

$$\mathbf{x}(k+1) = \mathbf{A}(k)\mathbf{x}(k) + \mathbf{B}(k)\mathbf{u}(k)$$

$$\mathbf{y}(k) = \mathbf{C}(k)\mathbf{x}(k) + \mathbf{D}(k)\mathbf{u}(k)$$

  where
  - $k \in \mathbb{Z}$ (integers)
  - signals x, u and y are number sequences

- A discrete-time system is a first order vector *difference equation*.

Broad Classes of Signals and Systems

- Signals:
  - Deterministic
  - Stochastic
    - Stationary
    - Non-stationary
- Systems
  - Linear and Non-linear systems
    - Time-Invariant Systems
    - Time-varying (Dynamical)
  - Continuous-Time (CT) and Discrete-Time (DT) systems
  - Univariate (SISO) vs Multivariate (MIMO) systems

AI 321: Discrete Linear Dynamical MIMO Systems with Stochastic Stationary (Theory)

Discrete Non-linear Dynamical Multivariate Non-stationary Systems (Practice)

References

[1] G. E. P. Box and G. Jenkins, *Time Series Analysis, Forecasting and Control*, 4th ed. USA: John Wiley and Sons, 2008, 2008, ISBN: 9780470272848.

[2] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing, Principles, Applications and Algorithms*, 3rd ed. USA: Prentice Hall Inc, 1996.

[3] R. A. DeCarlo, *Linear Systems, A State Variable Approach*, 1st ed. USA: Prentice Hall Inc, 1989.